

Chapter 6. Assist-Control Ventilation

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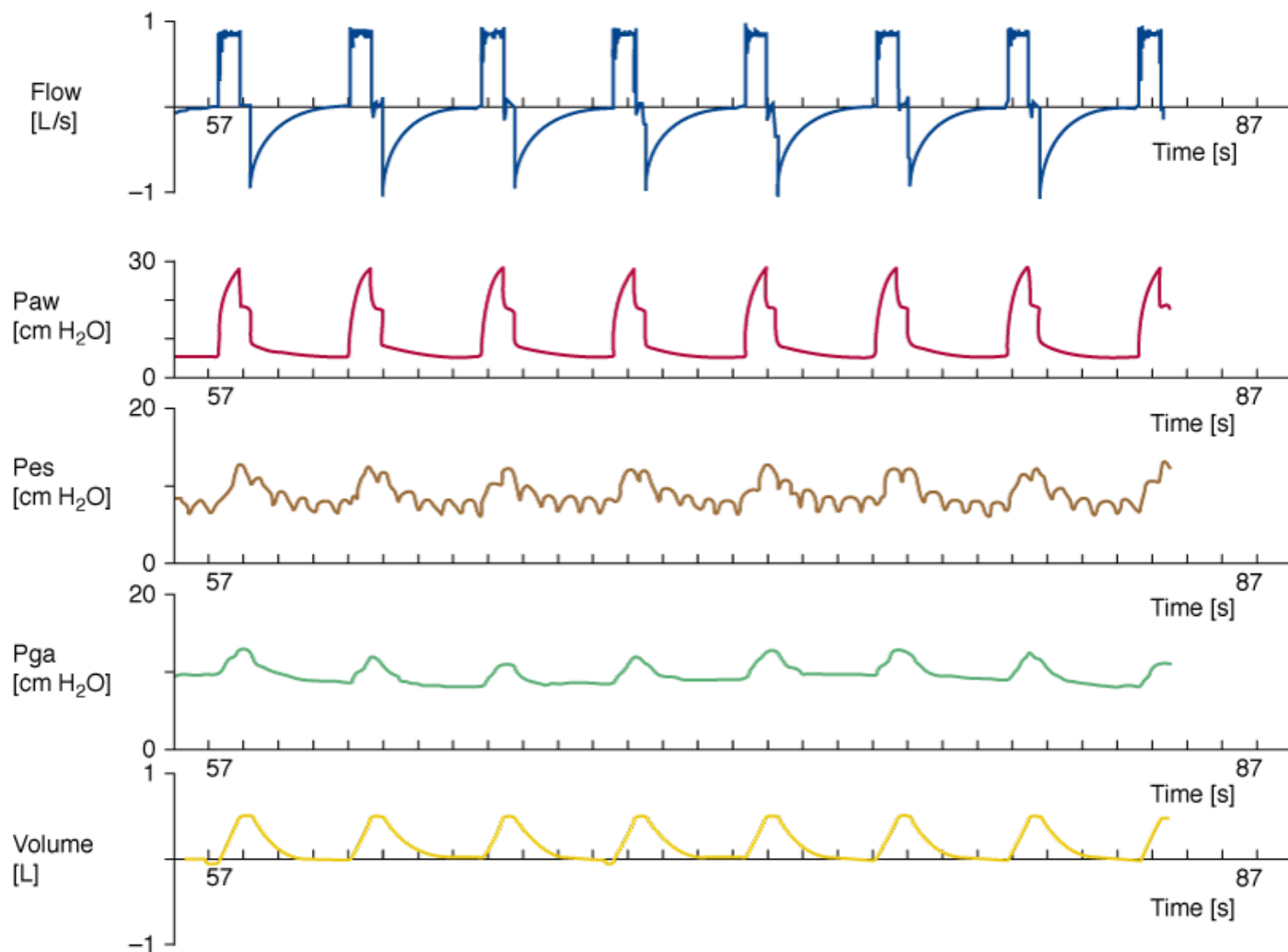
Assist-Control Ventilation: Introduction

Volume assist-control ventilation (ACV) is a ventilator mode in which the machine delivers the same tidal volume during every inspiration, whether initiated by the ventilator or by the patient. This occurs regardless of the mechanical load on the respiratory system and no matter how strenuous or feeble the inspiratory muscle effort. Current data indicate that ACV is still the most frequently used mode in intensive care units (ICUs).¹ Nowadays, the main reason for patients being admitted to an ICU is the need for mechanical ventilation,² and the most common reason to initiate mechanical ventilation is acute respiratory failure.^{1,3,4} Approximately 60% of intubated, ventilated patients receive ACV.⁵ This percentage is similar for patients ventilated for decompensated chronic obstructive pulmonary disease (COPD),⁵ and even higher for those ventilated for acute respiratory distress syndrome (ARDS).⁶

Basic Principles

In ACV, mechanical breaths can be triggered by the ventilator or the patient. With the former, triggering occurs when a certain time has elapsed after the previous inspiration if the patient fails to make a new inspiratory muscle effort ([Fig. 6-1](#)). The frequency at which time triggering takes place is determined by the backup rate set on the ventilator. When patients trigger a mechanical breath, their spontaneous inspiratory effort is sensed by the machine, usually as a change in airway pressure or airflow. When such a change crosses the trigger-sensitivity threshold, the ventilator delivers the preset tidal volume. [Chapter 3](#) provides a detailed explanation regarding the working principles of ventilators.

Figure 6-1



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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), esophageal pressure (*Pes*), gastric pressure (*Pga*), and tidal volume (*VOLUME*). Each mark on the time axis denotes 1 second. These recordings were obtained in a passively ventilated patient. Each breath is time-triggered.

Mechanical breaths have precise mechanisms for being initiated (trigger variable), sustained (limit variable), and stopped (cycle variable). These are known as phase *variables*.⁷ In ACV, the mechanical breaths are limited by volume and/or flow and cycled by volume or time. The inspiratory flow-shape delivery is usually a square (constant) during ACV, although some ventilators also permit sinusoidal and/or ramp (ascending or descending) gas flows.

Physiologic Effects

Mechanical ventilation is a lifesaving supportive treatment that improves gas exchange and decreases the mechanical workload of the respiratory muscles while buying time for the patient to recover. The way mechanical ventilation is used is central to its short-term and long-term effects. Ventilator settings are a major determinant of the physiologic and clinical effects of ACV. [Chapters 36](#) and [37](#) address the physiologic effects of ACV on gas exchange and cardiovascular function.

In every assisted mode, the ventilator responds to a patient's inspiratory effort. Both pressure-triggering and flow-triggering systems of modern ventilators offer high performance, and the differences are small in terms of the added work of breathing. A bench study comparing the performance of new-generation ventilators versus old-generation ventilators revealed that

triggering function, pressurization capacity, and expiratory resistance are globally similar, thus suggesting that a technological ceiling has been reached.⁸

Inspiratory Muscle Effort

Marini et al⁹ reported that decreases in trigger sensitivity increased the work of breathing. Although decreasing the inspiratory flow rate from 100 to 80 L/min did not affect the effort to breathe at a moderate minute ventilation (12 L/min), it increased the work expenditure significantly when the minute ventilation was doubled.⁹ When inspiratory flow was reduced to 40 L/min and thus did not match the subject's demand, the work of breathing increased by 50%. Marini et al¹⁰ later analyzed the inspiratory work at two inspiratory flow settings, 60 and 100 L/min, in twenty patients. Tidal volume was unchanged. The patients' work per liter of ventilation at both ACV inspiratory flow settings represented approximately 60% of the work dissipated during spontaneous breathing. Dead space was added in half the patients and led to marked increases in muscle effort. Patients' work of breathing did not correlate with minute ventilation, although it was highly correlated with respiratory drive and muscle strength. A decrease in inspiratory flow to 40 L/min (in five patients) led not only to an increase in effort but also to premature expiratory efforts, encroaching on the ventilator's inspiratory time. In total, these data demonstrated that inspiratory muscle effort persists throughout the inflation and that a substantial amount of muscle work is dissipated during ACV.

Ward et al¹¹ analyzed the inspiratory muscle effort at several inspiratory flow rates between 25 and 65 L/min and confirmed the findings by Marini et al. Cinella et al¹² compared ACV and assist pressure-controlled ventilation (PCV) and used two tidal volume settings (12 and 8 mL/kg) with an inspiratory time of 1 second and no end-inspiratory pause. At high tidal volumes, no differences were observed between the two modes in terms of breathing pattern or indices of inspiratory muscle effort. When a tidal volume of 8 mL/kg was used (and thus inspiratory flow decreased with ACV), differences arose. Respiratory rate and occlusion pressure (a measure of respiratory motor output) tended to be higher with ACV, and indexes of inspiratory muscle effort showed a marked increase as compared with PCV. In the second part of this study,¹² the authors compared the effects of ACV and assist PCV using a fixed tidal volume (8 mL/kg) and two different settings for inspiratory time: 1 second and 0.6 second with no pause. With the longer inspiratory time, the differences were the same as stated previously. When inspiratory time was reduced and thus inspiratory flow increased during ACV, the differences between the modes virtually vanished. The study showed that both modes unloaded the respiratory muscles equally, provided that the inspiratory flow rate was set appropriately during ACV. These data confirm the importance of maintaining an inspiratory flow rate high enough to satisfactorily unload the respiratory muscles and also point out that moderate to low tidal-volume ventilation using high flow rates results in a short inspiratory time, which may not be optimal for some patients. The duration of diaphragmatic contraction was unaffected by the ventilator settings and always was shorter than 1 second.¹² Thus, it appears that the effects of high airflow settings on muscle unloading are mainly exerted at the very beginning of inspiratory efforts. Similar results were obtained by McIntyre et al¹³ when comparing ACV with a pressure-limited volume-guaranteed dual mode. These authors, however, suggested that the pressure-limited breaths could reduce patient-ventilator flow dyssynchrony. These physiologic effects were confirmed in a subsequent study performed by the same group.¹⁴

Inspiratory Flow Settings and Breathing Pattern

A number of investigators have shown that patients^{12,15-17} and healthy individuals^{18,19} react to an increase in inspiratory flow with an increase in respiratory rate when tidal volume is kept constant. In these circumstances, the imposed ventilator inspiratory time shortens as flow increases. This leads to a decrease in neural inspiratory time. When tidal volume is increased by lengthening the duration of inspiratory flow, neural expiratory time increases, and respiratory rate tends to decrease. These changes have opposite effects on respiratory rate. The mechanisms explaining these responses are complex and include the

Hering-Breuer reflex (inhibits inspiration and prolongs expiration), reflexes mediated by vagal mechanoreceptors, and perhaps consciousness-mediated reflexes.^{20–22}

Airflow-induced changes in breathing pattern carry important clinical implications, especially in patients with dynamic hyperinflation. Because inspiratory time is made up of the time of flow delivery and inspiratory pause, Laghi et al¹⁷ hypothesized that a decrease in ventilator inflation time would cause an increase in rate. In ten noninvasively ventilated stable patients with an obstructive airway disease, the investigators increased flow at constant tidal volume and decreased the inspiratory pause, keeping inspiratory flow and tidal volume constant. When inspiratory time was decreased (by increasing flow from 30 to 90 L/min), respiratory rate and expiratory time increased significantly. Intrinsic positive end-expiratory pressure (PEEP) diminished significantly despite the increase in respiratory rate. When inspiratory time was decreased by shortening the inspiratory pause, both respiratory frequency and expiratory time increased significantly. Again, intrinsic PEEP decreased significantly. Additionally, the higher inspiratory flow rates also decreased respiratory drive and inspiratory effort. These results suggest that imposed ventilator inspiratory time duration determines the respiratory rate and that the strategies that reduce ventilator inspiratory time, although accompanied by an increase in respiratory rate, also prolong the time for exhalation, thus decreasing intrinsic PEEP.

Respiratory Muscles

Mechanical ventilation per se can induce respiratory muscle damage,^{23–25} and patients appear to exhibit diaphragmatic weakness after a period of mechanical ventilation.²⁶ The term *ventilator-induced diaphragm dysfunction* was coined to express the decrease in the force-generating capacity of the diaphragm that results after a period of passive controlled mechanical ventilation.²⁷ Le Bourdellès et al²⁸ showed that anesthetized, passively ventilated rats had lower diaphragmatic weight and a reduction in their force-generating capacity in comparison with spontaneously breathing control animals. Anzueto et al²⁹ studied sedated, paralyzed baboons under ACV for 11 days. Endurance time decreased over this period, and transdiaphragmatic pressure diminished by 25%, suggesting that the duration of passive ACV is also a relevant factor.

Sassoon et al³⁰ showed that 3 days of passive ventilation in rabbits led to a progressive decrease in the force-generating capacity of the diaphragm in comparison with control animals who received the same total amounts of sedatives but were breathing spontaneously. They also showed that significant diaphragmatic myofibril damage had occurred. Other authors have reported similar data.^{31,32} Several investigators^{33–36} have begun to elucidate the complex cellular, molecular, and gene expression mechanisms underlying passive ventilation-induced respiratory muscle damage. Such mechanisms include, among others, decreased protein synthesis, increased proteolysis, oxidative stress, and alterations in cytosol calcium metabolism.²⁷

Subsequent findings by Sassoon et al³⁷ carry important clinical implications. The authors found that ACV, as compared with passive ACV, can attenuate markedly the decrease in diaphragmatic force induced by total inactivity in rabbits. Another investigation with clinical ramifications has shown that passive ACV improves diaphragmatic force production in rats challenged with intravascular endotoxin as compared with equally challenged spontaneously breathing animals.³⁸

An interesting issue is the combined effects of certain drugs (e.g., corticosteroids) on diaphragmatic function during passive ventilation. Maes et al³⁹ studied the effects of corticosteroid administration in rats (a single injection of 80 mg methylprednisolone/kg) on diaphragm function. The animals were ventilated with passive ACV for 24 hours. The main finding of this investigation was that a very high dose of corticosteroids protected the diaphragm against the deleterious effects of passive ACV. The diaphragm of treated animals maintained force, fiber dimension, and myogenin protein levels, whereas the diaphragm of nontreated animals exhibited a reduction in force, fiber atrophy, and reduced myogenin expression. The mechanism of this protective effect is the avoidance of muscle proteolysis, probably mediated by calpain. In a similar study

with rabbits, Sassoon et al⁴⁰ showed that very high doses of methylprednisolone (60 mg/kg/day for 2 days) have no additive effects on diaphragmatic dysfunction induced by passive ACV. The same doses administered during ACV, however, produced a significant decrease in the maximal tetanic force elicited by the diaphragm.⁴⁰ Thus, the effects of high-dose methylprednisolone on diaphragmatic function depend on the mode of ventilation: if the muscle contracts, the effects are injurious, whereas if the muscle is passive, the effects are protective or neutral.

Important data have appeared in the last few years regarding the effects of ACV in human subjects. Levine et al⁴¹ showed that complete diaphragmatic inactivity for 18 to 69 hours (mean: 34 hours) in brain-dead subjects resulted in marked atrophy of slow and fast-twitch fibers of the diaphragm as compared to matched controls (individuals who were ventilated for a scheduled surgery for 2 to 3 hours). The major mechanism explaining the diaphragmatic atrophy was increased muscle proteolysis. Peripheral skeletal muscles (pectoralis major) did not show histologic findings of atrophy. A subsequent study by Hussain et al,⁴² conducted in humans and using a similar design, extended the findings and provided data suggesting that both protein synthesis and breakdown are involved in the diaphragmatic dysfunction.

Jaber et al⁴³ have investigated the time course of the decrease in diaphragm contractility in humans under passive ACV. The authors observed a progressive loss in diaphragmatic force, as reflected by measurements of tracheal pressure. The tracheal twitch pressure significantly decreased over time: mean reduction was 32% after 6 days of passive ventilation. The degree of muscle injury as detected by electron microscopy was significantly correlated with the duration of passive ACV. Again, upregulation of proteolytic systems played a major role in the ventilator-induced diaphragmatic injury induced by mechanical ventilation.

If passive ventilation is one extreme, the other is a fatiguing loading. Both extremes are harmful to the respiratory muscles. Normal subjects submitted to inspiratory-resistive loading up to a fatiguing threshold showed a decrease in diaphragmatic contractility lasting for at least 24 hours.⁴⁴ Jiang et al⁴⁵ showed diaphragmatic injury and inflammation at 3 days after a 90-minute period of acute moderate and high inspiratory-resistive loading in rabbits. The same group⁴⁶ subsequently reported a marked decrease in the force production of the diaphragm at 3 days after high inspiratory-resistive loading over the same time. Such stress also induces selective upregulation of a number of cytokines in the diaphragmatic fibers, and eventually may lead to systemic effects.^{47,48} Toumpanakis et al⁴⁹ have further analyzed the effects of inspiratory resistive breathing in rat lungs. The animals received 100% oxygen during the experiments. The authors showed that after 3 to 6 hours of stressful breathing, the alveolar-capillary membrane permeability increased, the static lung compliance decreased, and significant lung inflammation developed, as manifested by changes in histology (appearance of interstitial and intraalveolar neutrophils) and cytokine expression (increase in tumor necrosis factor and interleukin levels in lung tissue).

Sleep

Research studies conducted in patients admitted to an ICU reveal that patients experience major sleep disturbances in terms of quantity and quality.^{50–52} The acuity of illness, the use of medications (such as sedatives or opioids), caregiver interventions, and environmental elements are contributing factors.⁵⁰ Gabor et al⁵³ indicated that only 30% of sleep disruption in ventilated patients was attributable to elements of the ICU environment.

Parthasarathy and Tobin⁵⁴ sought to determine if sleep quality was influenced by the mode of ventilation. They hypothesized that sleep is more fragmented during pressure-support ventilation (PSV) as compared to ACV because of the development of central apneas. Eleven patients were ventilated with ACV at tidal volumes of 8 mL/kg, inspiratory flow rate 1 L/s, and a backup rate of four breaths below the total assisted rate. PSV was set to deliver the same tidal volume. Patients also received PSV with 100 mL of added dead space. During wakefulness, respiratory rate was similar with the two modes. During sleep, minute ventilation fell more during PSV than during ACV. Sleep fragmentation, measured as the number of arousals and awakenings,

was significantly greater during PSV than during ACV (seventy-nine versus fifty-four events per hour). Six patients had apneas while receiving PSV, whereas none had apneas while receiving ACV. The percentage of patients who had congestive heart failure was significantly higher among patients exhibiting apneas than among patients free of apneas (83% vs. 20%). Minute ventilation during sleep was greater in patients who did not develop apneas, suggesting that increased drive protects against the development of apneas. The addition of dead space reduced the number of apneas markedly: from fifty-four to four apneas per hour. These data suggest that settings that generate overassistance promote the occurrence of apneas during assisted ventilation.

Cabello et al⁵⁵ conducted a study in fifteen ventilator-dependent patients and compared three modes: ACV, PSV, and automatically adjusted PSV. The hypothesis was that PSV settings adjusted to patient ventilatory needs could improve sleep quality as compared to ACV. During ACV, settings were adjusted to provide a tidal volume of 8 mL/kg with a constant inspiratory flow of 1 L/s (and backup rate at 10 breaths/min). In the second arm, PSV was adjusted by clinicians to obtain a tidal volume of 6 to 8 mL/kg and a respiratory rate below 35 breaths/min. In the third arm, PSV was automatically regulated in a way that continuously adjusted the level of support so as to keep the patients within a comfort zone.⁵⁶ PEEP was kept constant at 5 cm H₂O, and patients were free of sedative drugs. The median tidal volumes (390 to 500 mL), respiratory rates (20 to 21 breaths/min), and minute ventilation did not differ between the three modes. Nine patients exhibited sleep apneas, and ten displayed ineffective efforts. The number of ineffective efforts per hour of sleep did not differ among the modes (mean: six to sixteen ineffective efforts per hour). The number of apneas was similar between the two PSV modalities (five to seven apneas per hour of sleep). Sleep fragmentation (arousals and awakenings per hour), sleep architecture, and sleep quantity did not differ among the modes. One explanation for the difference with the findings in the study by Parthasarathy and Tobin is that tidal volumes and minute ventilation were similar for all modes in the study of Cabello et al. The relative infrequency of ineffective efforts and apneas in this study suggests that patients were not overassisted. Together these data indicate that excessive ventilator support is central in the development of sleep fragmentation.

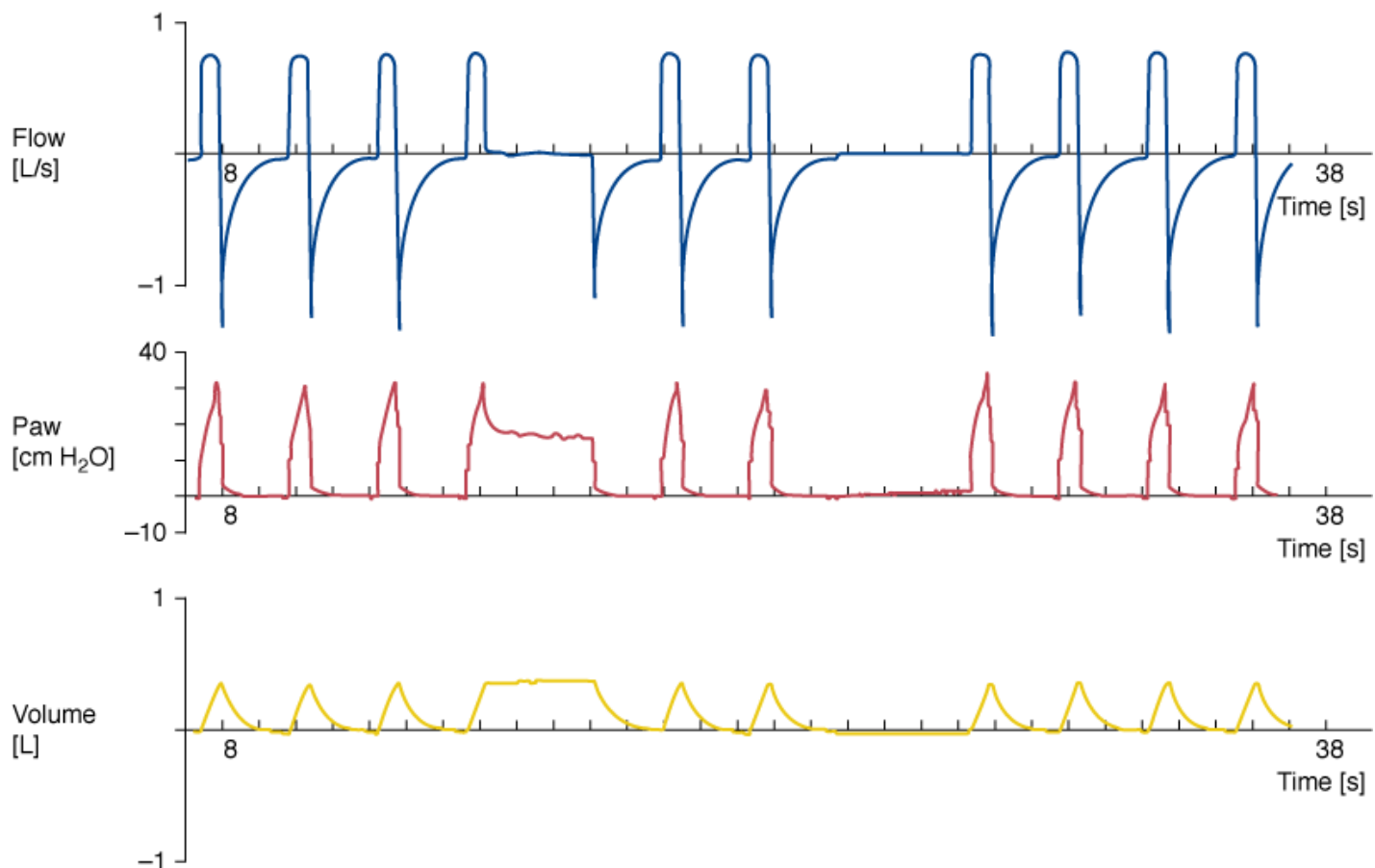
The clinical consequences of these sleep abnormalities are not known. Researchers have noted that sleep deprivation may generate immune suppression, loss of circadian hormonal secretion (melatonin and cortisol), profoundly alter respiratory muscles endurance and neurocognitive function, and modify the normal physiologic responses to hypoxia and hypercapnia.^{51,52} Whether the sleep disturbances are a marker of brain dysfunction related to critical illness or represent a specific syndrome with an independent effect on outcomes is not known.

Rationale, Advantages, and Limitations

The main reasons for using ACV are to unload the inspiratory muscles and to improve gas exchange. ACV permits complete respiratory muscle rest, which is usually the case when patients do not trigger the machine, and a variable degree of respiratory muscle work. ACV commonly achieves an improvement in gas exchange, and only a minority of ventilated patients die because of refractory hypoxemia.

During passive ventilation with ACV at a constant inspiratory flow, fundamental variables related to respiratory system mechanics, such as tidal volume, inspiratory flow, peak airway pressure, end-inspiratory plateau airway pressure, and total PEEP (the sum of external PEEP and intrinsic PEEP, if any), are measured easily (Fig. 6-2). These variables allow calculation of resistance, compliance, and the time constant of the respiratory system.

Figure 6-2



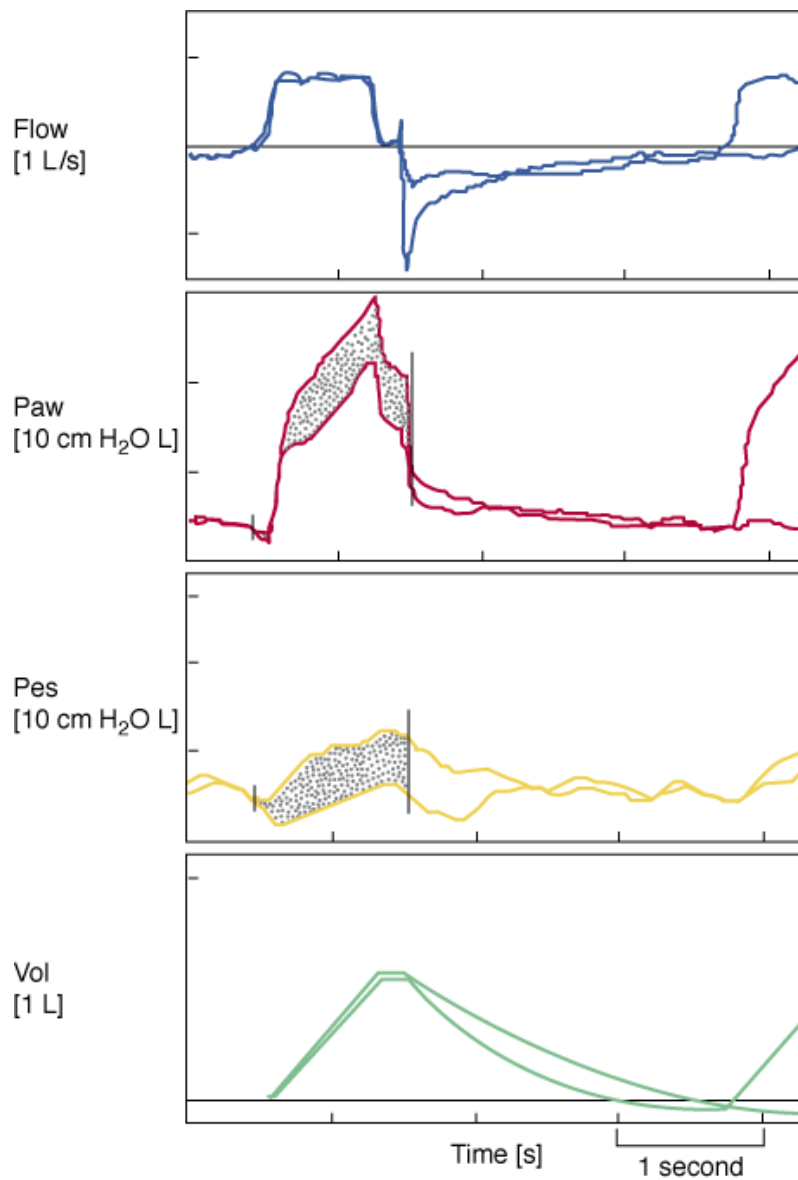
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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), and tidal volume (*VOLUME*). Each mark on the time axis denotes 1 second. Note that expiratory flow is interrupted by the beginning of each breath, thus heralding dynamic hyperinflation. A prolonged end-inspiratory occlusion (fourth breath from the left) enables measurement of the static recoil pressure of the respiratory system. A prolonged end-expiratory occlusion (sixth breath from the left) illustrates the presence of intrinsic PEEP (3 cm H₂O). These values, together with peak airway pressure, inspiratory flow rate, and tidal volume, enable the calculation of resistance, compliance, and respiratory system time constant in passively ventilated patients.

If airway pressure tracings are obtained during passive ACV as well as during patient-triggered ACV at the same settings, we can estimate a patient's work of breathing simply by superimposing the two tracings (Fig. 6-3). When patients are triggering the breaths, the end-inspiratory plateau pressure also can be influenced by the amount and duration of inspiratory muscle effort (Fig. 6-4). When mechanical breaths are triggered by the patient, the scooping on the airway pressure profile allows indirect evaluation of patient-ventilator interaction (Fig. 6-5). Such capabilities are unique to ACV. These capabilities represent a major advantage because they enable one to properly understand respiratory system mechanics and patient-ventilator interactions.

Figure 6-3



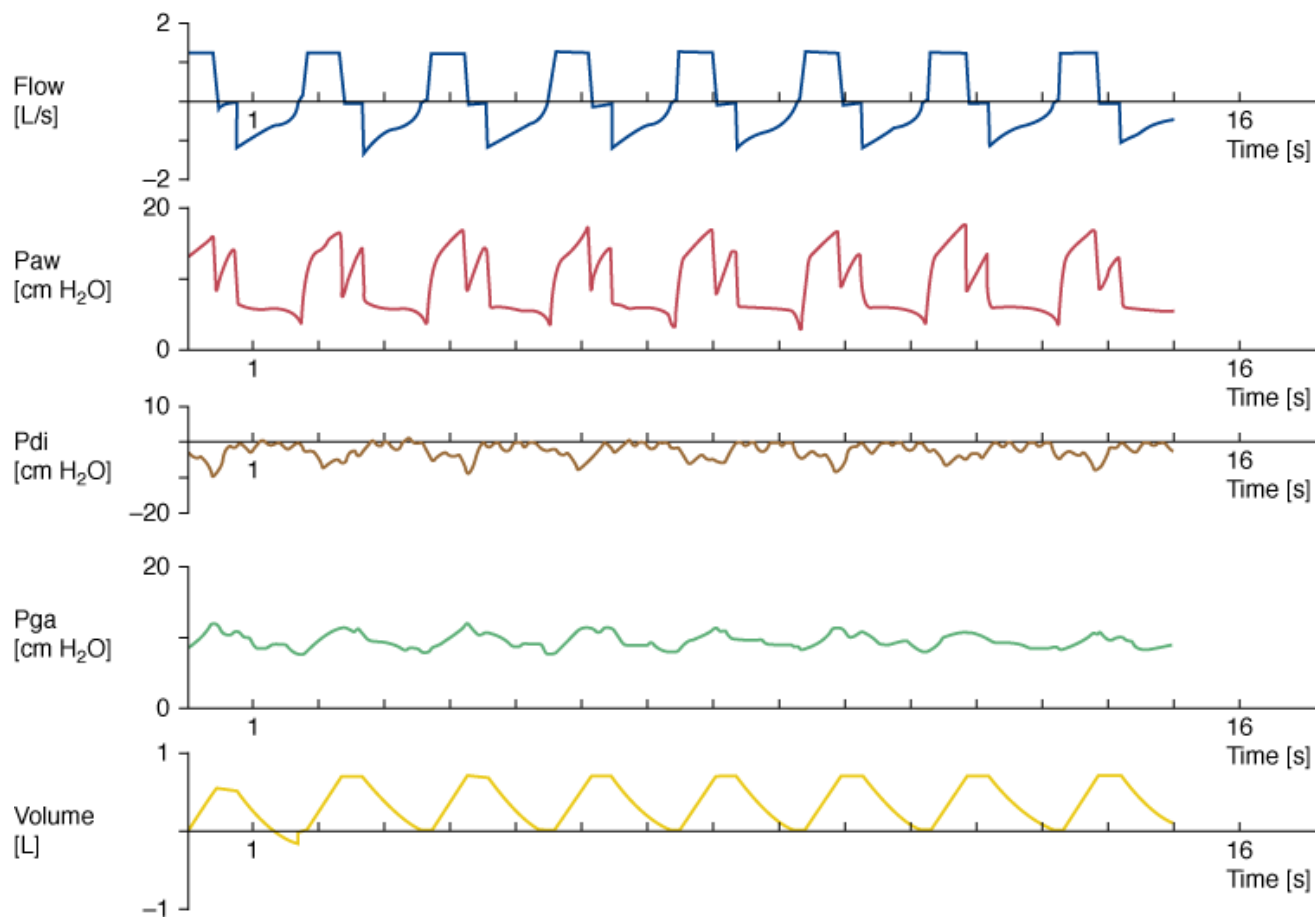
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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), esophageal pressure (*Pes*), and tidal volume (*VOL*).

Each mark on the time axis denotes 1 second. The tracings in each panel were obtained from the same individual at two different times. They were then superimposed. Ventilator settings were identical. The vertical line on the airflow, airway pressure, and esophageal pressure tracings indicates the end of ventilator's total inspiratory time. The *dotted areas* within the airway pressure and esophageal pressure tracings are identical. The *dotted areas* denote the amount of inspiratory muscle effort that the patient made during the assisted breath.

Figure 6-4

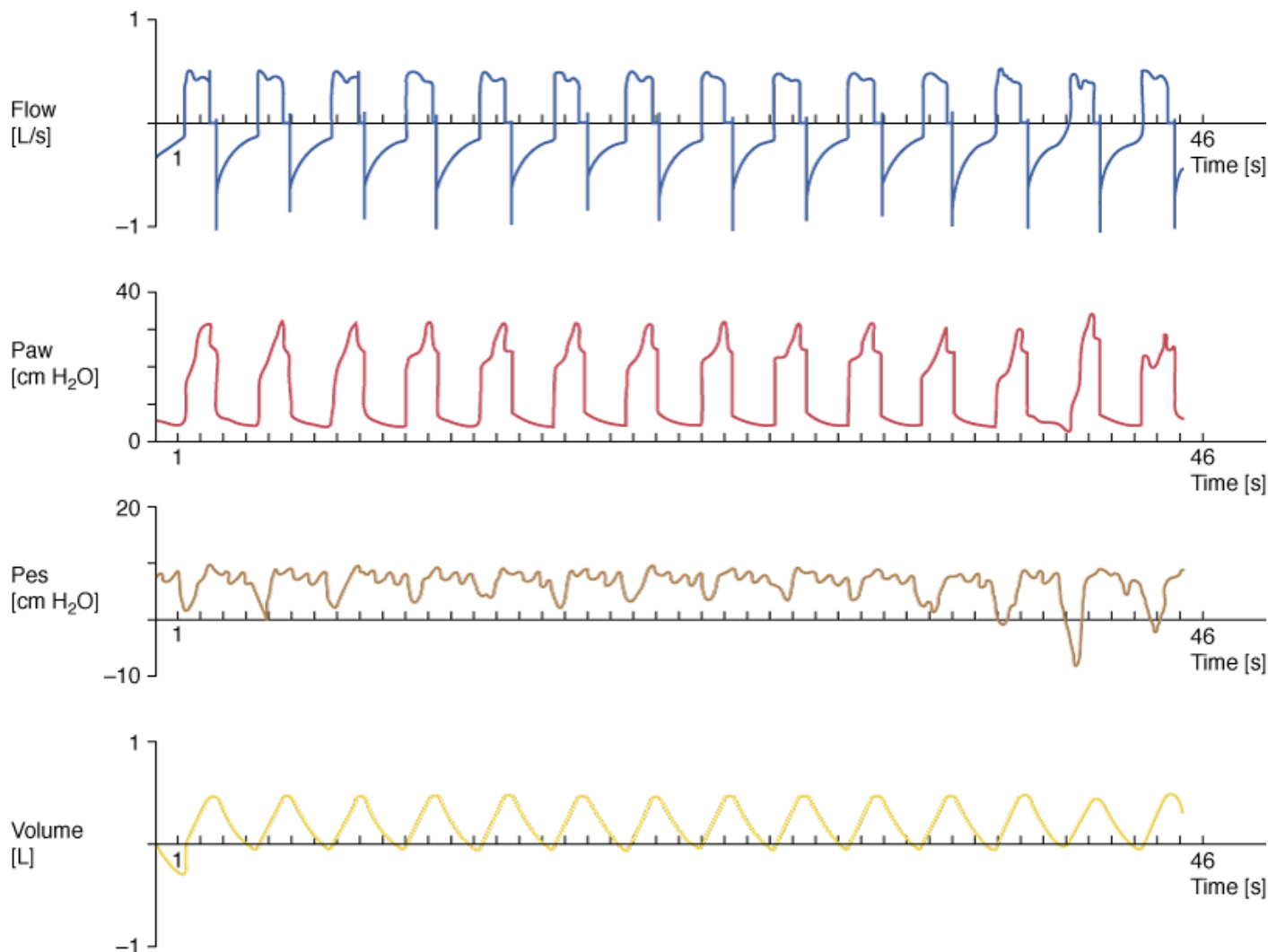


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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), transdiaphragmatic pressure (*Pdi*), gastric pressure (*Pga*), and tidal volume (*VOLUME*). Each mark on the time axis denotes 1 second. The time of inspiratory flow is shorter than the duration of diaphragmatic contraction. This patient did not exhibit expiratory muscle recruitment, as indicated by the gastric pressure recording. Inspiratory muscles relax at the end of the inspiratory pause time, thus explaining the “M” wave shape on the airway pressure recording.

Figure 6-5



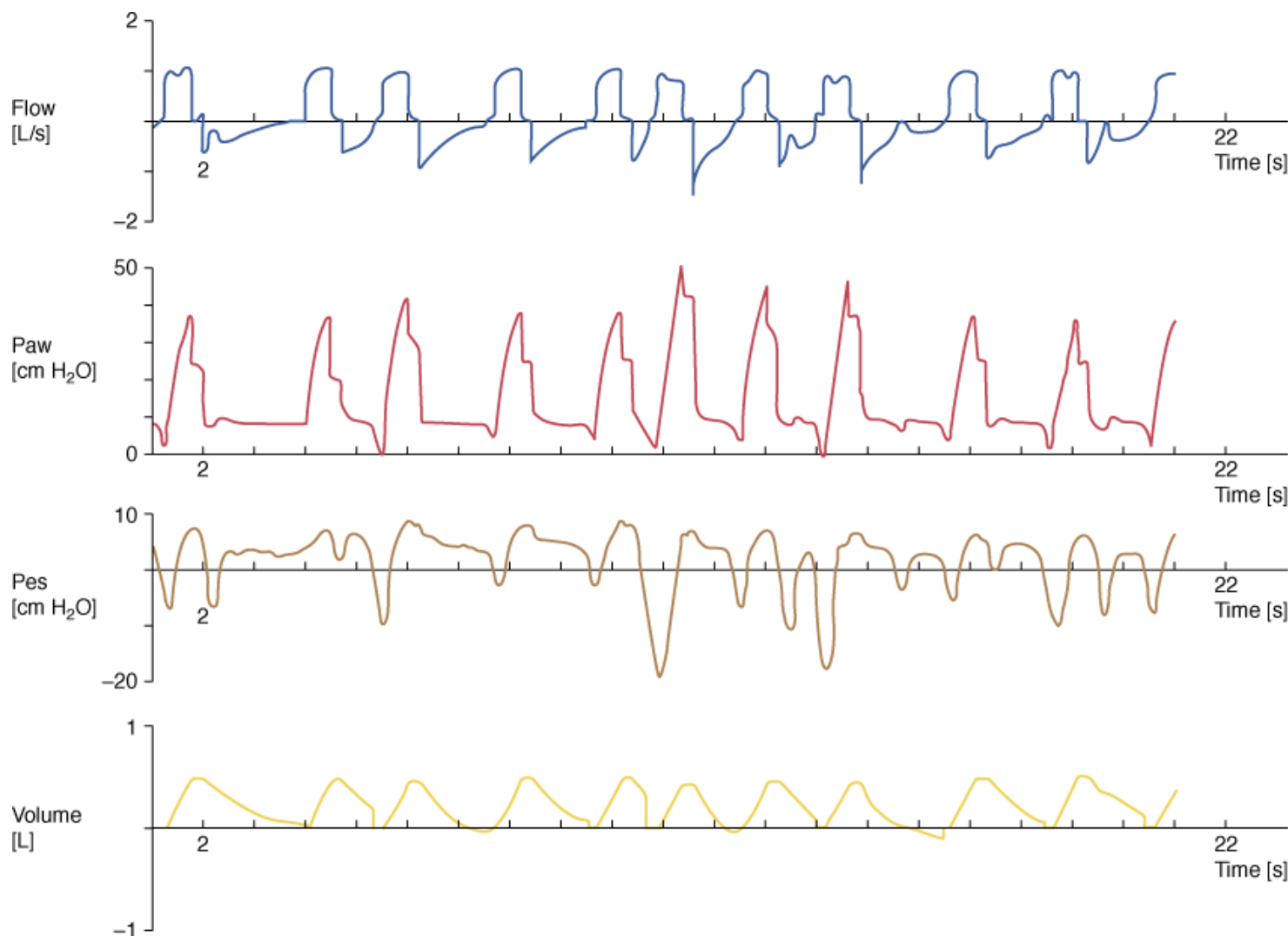
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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), esophageal pressure (*Pes*), and tidal volume (*VOLUME*). Each mark on the time axis denotes 1 second. As seen easily on the esophageal pressure tracing, there is a highly variable inspiratory muscle effort over time, thus inducing permanent scooping on the airway pressure recording.

A major limitation of ACV is that it imposes a number of constraints on the variability of the patient's breathing pattern: inspiratory flow, inspiratory time, and backup rate. Adjusting ACV settings may be more complex than with pressure-limited modes. One reason is that manufacturers employ different algorithms for implementing the delivery of a tidal breath. The other reason is that during ACV it is difficult to pinpoint the inspiratory flow rate and tidal volume settings that are optimal for an individual patient. Some settings are almost impossible to achieve with ACV. For instance, the simultaneous adjustment of a moderate tidal volume at a high inspiratory flow rate will produce a short machine inspiratory time, which, under certain circumstances, may not match the patient's neural inspiratory time properly. In addition, the patient's varying ventilatory needs and the change in the mechanical properties of the respiratory system over the course of ventilation imply that periods of underassist are likely to be interspersed with periods of overassist (Fig. 6-6). These problems, however, are common to most ventilator modes.

Figure 6-6



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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), esophageal pressure (*Pes*), and tidal volume (*VOLUME*). Each mark on the time axis denotes 1 second. Note the marked breath-by-breath variability in inspiratory muscle effort (esophageal pressure swings) and airway pressure profile. There is profound patient-ventilator dyssynchrony, and numerous inspiratory attempts fail to trigger the ventilator. It is also remarkable how difficult it can be to estimate the end-inspiratory plateau airway pressure in such circumstances.

Indications and Contraindications

ACV is indicated when a life-threatening physiologic derangement in gas exchange or cardiovascular dynamics has not been corrected by other means. Clinical manifestations of severely increased work of breathing or impending respiratory arrest are indications for instituting ACV.⁵⁷ Although there appear to be no absolute contraindications to ACV, some of its shortcomings may prompt physicians to use other modes.

Comparison with Other Modes

Pressure-Controlled Ventilation, Airway Pressure Release, and Adaptive Support Ventilation in Acute Respiratory Failure Patients

During PCV, the ventilator functions as a pressure controller, and operates in a pressure-limited and time-cycled mode. With PCV, delivery of airflow and tidal volume changes according to the mechanical impedance of the respiratory system and patient inspiratory muscle effort. This mechanism implies that every increase in transpulmonary pressure is accompanied by an increase in tidal volume. Numerous studies^{58–68} have compared the effects of PCV and ACV. In general, these studies included a limited number of patients and different adjustments were used. Taken together, no major differences in terms of gas exchange and major outcomes emerge between ACV and PCV.

Two studies^{69,70} have analyzed outcomes between ACV and airway pressure release ventilation (APRV). APRV is similar to PCV except that it allows spontaneous breathing (in the form of continuous positive airway pressure) at any part of the ventilatory cycle. The study by Maxwell et al⁷⁰ was carried out in trauma patients: thirty-two received ACV and thirty-one were ventilated with APRV. Tidal volumes did not differ between the modes and, although patients ventilated with APRV had higher mean airway pressures, the $\text{PaO}_2/\text{FiO}_2$ ratios were similar. Days of mechanical ventilation, ICU length of stay, incidence of pneumothoraces, need for tracheostomy, and ventilator-associated pneumonia rates did not differ between the two modes. Sedative doses were similar between the modes and mortality was almost identical (approximately 6.4% in each group).

González et al⁶⁹ compared outcomes of assorted patients receiving ACV ($n = 234$) and APRV ($n = 234$) as a primary mode. The data were obtained from an observational multicenter cohort study, and a case-matched analysis on propensity score was performed. No differences in relevant clinical outcomes were detected between the modes: days of mechanical ventilation, ICU and hospital length of stay, reintubation rates, and hospital mortality. Surprisingly, the tracheostomy rate was significantly higher in the patients who were ventilated with APRV (20%) versus ACV (11%). Tidal volumes were similar (approximately 9 mL/kg) in both groups. The partial pressure of arterial oxygen (PaO_2)-to-fractional inspired oxygen concentration (FiO_2) ratio was significantly higher with APRV than with ACV (263 vs. 232 mm Hg), probably because expiratory pressure was lower with ACV (3 vs. 7 cm H_2O). The dominant factor associated with the use of this mode was geography: 196 of 234 patients receiving APRV were located in German ICUs.

Chung et al⁷¹ compared high-frequency percussive ventilation and ACV in patients with acute respiratory failure secondary to severe burns. High-frequency percussion is pressure-limited and time-cycled at high frequency (above 300 breaths/min) and is superimposed on a biphasic inspiratory and expiratory pressure cycle set at a normal rate (approximately 10 to 15 cycles/min). A total of sixty-two patients were randomized—thirty-one to ACV and thirty-one to percussive ventilation. Patients treated with ACV received fixed tidal volumes (6 mL/kg) and PEEP levels were set according to an algorithm. Mean airway pressure was virtually identical with the two modes and a significantly better PaO_2 -to- FiO_2 ratio was observed in the percussive ventilation group over the first 3 days. Sedation requirements and ventilator-free days did not differ between the groups. Although rescue ventilator therapy (29% vs. 6%) and barotrauma (13% vs. 0%) were more frequent during ACV as compared to percussive ventilation, no differences in mortality rates were observed between the two groups (19% each).

Adaptive-support ventilation is a closed-loop mode that selects a target ventilatory pattern based on patient weight, minimum minute volume, and a pressure limit. The ventilatory pattern is selected to minimize the total work of inspiration. Two studies have compared adaptive support ventilation with ACV. Sulemanji et al⁷² conducted a bench test with different experimental scenarios of lung mechanics, PEEP levels and body weights. Target volume during ACV was 6 mL/kg and inspiratory time was 0.8 second. Adaptive-support ventilation was able to maintain a lower plateau pressure compared to ACV in settings of low compliance, high PEEP, and high minute volume. This resulted from the expected decrease in tidal volume during adaptive-support ventilation when facing a condition of high impedance, especially low compliance. During adaptive support ventilation, the plateau pressure was exceeded by approximately 2 cm H_2O , whereas in ACV the plateau pressure was exceeded by up to 10 cm H_2O .

This behavior of adaptive-support ventilation has been also reproduced in passively ventilated patients.^{73,74} In the study of Iotti et al,⁷⁴ ACV (as set by the attending clinician) was compared to adaptive support ventilation (set to obtain the same minute ventilation). A total of eighty-eight patients were studied: twenty-two with no obvious lung disease, thirty-six with a restrictive disease, and thirty with obstructive disease. Adaptive support ventilation achieved a lower respiratory rate (17 vs. 19 breaths/min) and larger tidal volume than during ACV (9.4 vs. 8.4 mL/kg). Compared to ACV, adaptive support ventilation achieved a slight decrease in partial pressure of arterial carbon dioxide (Pa_{CO_2}) (from 41.6 to 40 mm Hg) and lower machine work (from 17.7 Joules/L to 14.6 Joules/L) when Pa_{CO_2} was kept constant. Compared to ACV, the obstructed patients received larger tidal volumes at lower respiratory rates. In restrictive patients, adaptive-support ventilation selected the lowest tidal volume (4.8 mL/kg) in patients with the shortest time constant (low compliance) and the highest tidal volume (10 mL/kg) in patients with normal time constant (near-normal compliance). These data suggest that adaptive support ventilation in passively ventilated patients is at least as efficient as ACV in terms of alveolar ventilation and work of breathing, and it is able to provide a breathing pattern tailored to the individual respiratory system mechanics.

Intermittent Mandatory Ventilation

Marini et al⁷⁵ compared ACV with intermittent mandatory ventilation (IMV) to provide 80%, 60%, 40%, 20%, and 0% of the ventilation observed during ACV (100% IMV). Tidal volume and flow settings during ACV were 10 mL/kg and 1 L/s, respectively, and average respiratory rate was 23 breaths/min. The total breathing frequency and spontaneous tidal volume increased as far as IMV assistance was decreased. Duration of inspiratory effort during assisted breaths was similar across the different IMV levels. At all levels of support, patients performed a substantial effort during the machine-assisted breaths that increased progressively as IMV assistance was withdrawn.

These data emphasize several points. Machine assistance does not suppress patient effort. There is a poor adaptation to ventilator assistance on a breath-by-breath basis, suggesting that the intensity of muscle effort is fixed before cycle initiation. Off-switching of inspiratory muscle contraction is independent of volume and flow ventilator settings. Viale et al⁷⁶ showed a rapid and gradual downregulation of inspiratory muscle effort and respiratory drive in ventilator-dependent patients with COPD, when they were switched from spontaneous unassisted breathing to PSV. This downregulation needed 6 to 8 breaths to achieve total stability, and the authors speculated that possible mechanisms explaining the gradual response were changes in Pa_{CO_2} and vagal stimulation.⁷⁶

Leung et al⁷⁷ compared ACV, IMV (80%, 60%, 40%, and 20% levels of assist), PSV (100%, 80%, 60%, 40%, and 20% levels of assist), and a combination of IMV with PSV of 10 cm H_2O . No PEEP was used. Average tidal volume and respiratory rate during ACV were 600 mL and 17 breaths/min, respectively. The observed rate during ACV was considered equivalent to IMV 100%. PSV 100% (average: 17 cm H_2O) was the level of assistance that resulted in the same tidal volume as during ACV; this led to a respiratory rate of 16 breaths/min. Nontriggering attempts occurred with every mode and were most numerous at the highest levels of assistance. At 100% levels of assistance, the inspiratory effort and dyspnea sensation were similar among the modes, and both increased progressively when assistance was decreased. When the total inspiratory effort was partitioned between its triggering and posttriggering components, the former was unchanged despite varying levels of ventilator assistance. The posttriggering effort, however, was highly correlated with the respiratory drive at the beginning of the breath.

Outcomes in patients receiving synchronized IMV with PSV ($n = 350$) and patients receiving ACV ($n = 1228$) as primary ventilator support modes have been compared.⁷⁸ Physicians were more likely to select synchronized IMV with PSV in less-sick patients, and ACV was mostly used in severely ill patients. After adjusting for a propensity score, no differences between the modes were detected in the duration of weaning, rates of reintubation, tracheostomy, or mortality. The clinical relevance of these data is not straightforward, as the propensity score only had a moderate level of discrimination.

Pressure-Support Ventilation

Tokioka et al⁷⁹ compared ACV with PSV set to achieve the same value of peak airway pressure as during ACV. This resulted in PSV levels of 27 cm H₂O above a PEEP of 12 cm H₂O. With these settings, tidal volume was significantly higher and machine respiratory rate significantly lower during PSV. These data indicate that peak airway pressure during ACV is an inappropriate surrogate variable to adjust PSV to get similar levels of assistance. Tejada et al⁸⁰ compared ACV with PSV in patients with respiratory failure of assorted etiologies. PSV was adjusted to deliver the same tidal volume as with ACV, although it actually resulted in significantly higher tidal volumes. The authors found a slightly better partial pressure of arterial oxygen (PaO₂) only in a subgroup of patients with restrictive disorders. Surprisingly, the calculated shunt in these patients (18% to 20%) was lower than in patients with COPD (26% to 29%). A significantly higher dead-space-to-tidal-volume ratio also was observed during PSV (24%) than during ACV (18%). These are extremely low values for patients with respiratory failure. For these reasons, the overall clinical significance of these findings is difficult to judge.

Kreit et al⁸¹ analyzed work of breathing during ACV and PSV in eleven patients. During ACV, tidal volume was 10 to 12 mL/kg, and inspiratory flow was 75 to 80 L/min. PSV was increased progressively to reach the same tidal volume as during ACV. This strategy resulted in an average pressure support of about 19 cm H₂O. The authors confirmed previous studies indicating that both work of breathing and respiratory rate vary inversely with the PSV level. With such adjustments, the patient work of breathing and minute ventilation were almost identical between the modes.

In a study not specifically designed to compare ACV with PSV, Aslanian et al⁸² used the average tidal volume measured during clinician-titrated PSV for later adjustments of ACV. PSV levels were set at a target respiratory rate of between 15 and 30 breaths/min, which resulted in an average PSV of 16 cm H₂O and an average tidal volume of 500 mL. Settings during ACV were tidal volume 500 mL with an inspiratory flow rate at 50 L/min. With such adjustments, respiratory rate, minute ventilation, breathing pattern, and several indexes of inspiratory muscle effort were similar between the modes.

Chiumello et al⁸³ compared the effects of PSV at 5, 15, and 25 cm H₂O with assist PCV at the same levels of pressure and inspiratory time as during PSV and ACV. ACV was delivered with a square and decelerating flow pattern, both matched for the same tidal volume and peak inspiratory flow as during PSV. No differences among the modes were observed. The authors also compared clinician-titrated PSV (average 10 cm H₂O) with two ACV modes (square and decelerating flow), both at two flow settings (high and low). Tidal volume was always the same. The peak inspiratory flow obtained during PSV (0.78 L/s) was the high-flow setting for both ACV types. When high-flow settings were used, no differences were observed. The low-flow setting, approximately 0.64 L/s, induced a significant increase in work of breathing without differences in respiratory rate or gas exchange.

In a selected population of patients with acute lung injury, Cereda et al⁸⁴ studied the physiologic changes that appeared during the 48 hours after the transition from ACV to PSV. Hemodynamics and oxygenation were similar. An increase in minute ventilation and a lower PaCO₂ were observed during PSV. Of forty-eight patients, ten did not tolerate PSV. These patients had a lower static compliance and a higher dead-space-to-tidal-volume ratio when compared with patients who succeeded. These data suggest that PSV might be an alternative to ACV in carefully selected patients with acute lung injury.

Biologically Variable Ventilation

Tidal volume during ACV is, by design, delivered in a monotonous manner. To re-create the spontaneous variability of physiologic rhythms, mechanical ventilation using computer-generated biologic variability in respiratory rate and tidal volume has been used. The goal of this approach is to improve gas exchange and respiratory system mechanics, and to minimize ventilator-induced lung injury. Data comparing ACV with ventilation achieved with randomly variable tidal volumes and

respiratory rates have been obtained in several experimental models of acute lung injury.^{85–91} Models include different animal species (e.g., rodents, pigs, and dogs), different type of insults (e.g., chemical, mechanical, and biologic), and different ventilator settings (e.g., PEEP or no PEEP). In these short-term experiments, variable ventilation was matched to ACV in terms of minute ventilation.

All studies, except the study by Nam et al,⁹⁰ reported benefits of variable ventilation over ACV in terms of arterial oxygenation, lung mechanics, redistribution of pulmonary blood flow, degree of lung edema, proinflammatory cytokine production, histologic damage, or combinations of these. Variable ventilation may induce a better distribution of tidal volume—thereby matching ventilation to perfusion—better recruitment, and increased surfactant production. Whether these putative benefits are attributable directly to the ventilator mode per se, the type of injury, the animal species, the ventilator settings (degree of variability, PEEP levels), or the respiratory system mechanical characteristics is unclear. The physiologic effects of this new mode in patients with acute lung injury are still unknown.

Variation in Delivery among Ventilator Brands and Troubleshooting

This section does not pretend to explain exhaustively the working principles of the dozens of different mechanical ventilators available on the market. The decision to stick with one style or another depends solely on the manufacturer. Some machines are user-configurable, but in different ways (inspiratory flow rate, inspiration-to-expiration ratio, and so on).

The fundamental settings during ACV are respiratory rate, tidal volume, and inspiratory flow rate. The backup respiratory rate determines the total breath duration, and both tidal volume and inspiratory flow rate determine the duration of mechanical inflation within a breath. The inspiratory pause, if used, appears immediately after the machine's flow delivery has ceased and thus increases the inspiratory time. The expiratory time is the only part of the breathing cycle that is allowed to vary when a patient triggers an ACV breath. For this reason, we consider machines that require inspiratory-to-expiratory ratio adjustment during ACV to be totally counterintuitive.

Some ventilators allow direct setting of respiratory rate, tidal volume, inspiratory flow rate, and inspiratory pause time. In my opinion, this is the most comprehensive approach, because the time for flow delivery depends on the tidal volume and inspiratory flow rate. Mechanical ventilators are lifesaving machines when used properly. Inappropriate use can be life-threatening. Because manufacturers follow different principles and strategies to build their machines, it is fundamental to get acquainted with the specifics of each ventilator and read the instruction manual carefully.

Recent bench studies^{8,92} evaluating the performance of multiple ventilators, have shown that triggering delay, pressurization capacity during PSV, tidal volume delivery during ACV, and expiratory resistance significantly differ across a wide range of new-generation ventilators. It is interesting to note that new turbine-based ventilators perform better (on average) in terms of trigger function and pressurization quality when compared to conventional servo-valve ventilators and perform as well as the best compressed-gas ICU ventilators.

Comparisons between target tidal volume and actually delivered tidal volume during ACV at different impedance conditions, and taking into account the differences in gas temperature and humidity between inspiration and expiration, showed marked differences across various new-generation ICU ventilators.⁹² Tidal volumes of 300, 500, and 800 mL were selected. Differences between targeted and delivered tidal volume ranged between –13% and +32%. Interestingly, ventilators that use compensation algorithms (which account for volume compensation when gas is compressed) delivered significantly larger tidal volumes (although less than 10% on average) than preset tidal volumes under body temperature and pressure-saturated conditions.⁹² The clinical relevance of these differences needs to be carefully evaluated.

Solving problems related to mechanical equipment requires special skills and intuition. Some troubles are intrinsically related to machines and their own working principles/algorithms but can be minimized if manufacturers' recommendations are

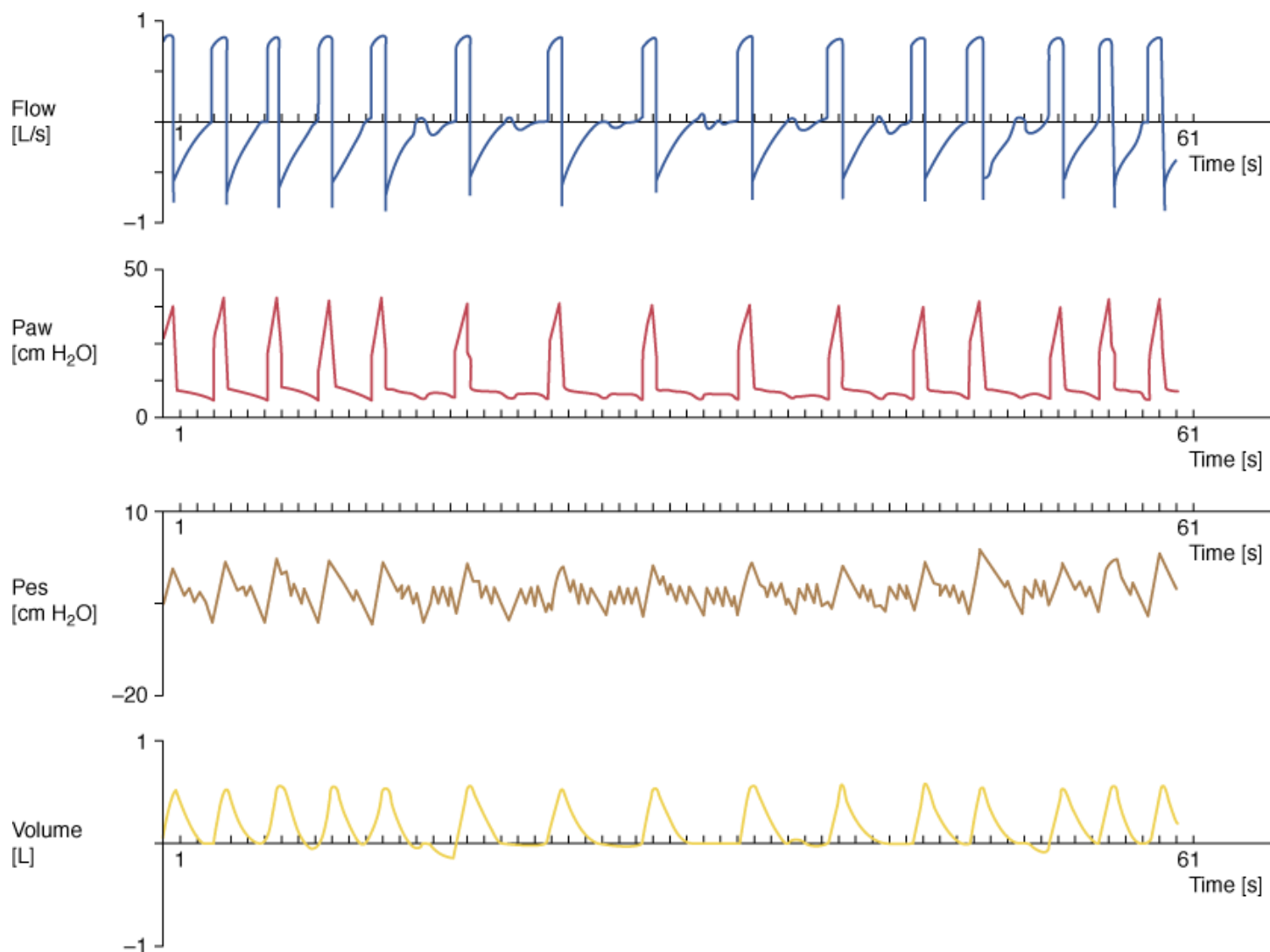
followed. It should be unnecessary to emphasize that thorough reading of the operator's manual is mandatory. Overall, the reported frequency of ventilator malfunctions seems to be very low.⁹³

Adjustments at the Bedside

Settings to be adjusted in ACV are inspired oxygen concentration, trigger sensitivity (to be set above the threshold of autotriggering), backup rate, tidal volume, inspiratory flow rate (or inspiratory time), end-inspiratory pause, and external PEEP, if any. When ACV is instituted after tracheal intubation, patients usually are sedated and passively ventilated. Proper measurement of end-inspiratory plateau airway pressure and calculations of compliance and airflow resistance may help in adjusting the ventilator's backup breathing pattern. The time constant of the respiratory system determines the rate of passive lung emptying. The product of three time constants is the time needed to passively exhale 95% of the inspired volume.^{94,95} If expiratory time is insufficient to allow for passive emptying, this will generate hyperinflation.

During ACV, when a patient triggers a mechanical breath, the expiratory time is no longer constant. Consequently, exhaled volume might change on a cycle-to-cycle basis and modify the degree of dynamic hyperinflation. This may alter patient-ventilator synchrony and cause subsequent wasted inspiratory efforts, as is seen in patients with low inspiratory drive (i.e., patients who are sedated) and those with prolonged time constants (Fig. 6-7). One study showed that sedation level is a predictor of ineffective triggering⁹⁶ and at least two studies showed that patient-ventilator asynchrony (mainly ineffective triggering) is associated with worse outcomes: increased duration of mechanical ventilation, more tracheostomies, and lower likelihood of being discharged home.^{97,98} Importantly, ineffective triggering is associated not only with sedatives and the presence of an obstructive disease, but also with excessive levels of support and excessive tidal volumes.⁹⁷⁻⁹⁹

Figure 6-7



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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(From top to bottom) Tracings of airflow (*FLOW*), airway pressure (*Paw*), esophageal pressure (*Pes*), and tidal volume (*VOLUME*). Each mark on the time axis denotes 1 second. As can be seen from the esophageal pressure recordings, this patient was markedly unloaded and exhibited a feeble respiratory drive. As a result, multiple wasted inspiratory efforts are interspersed between the patient-triggered breaths.

Chapters 29, 30, 31 discuss mechanical ventilation in specific scenarios. Some general principles, however, are worth recalling. The goals of mechanical ventilation, in particular during ACV, have changed profoundly in the last years. Nowadays, moderate tidal volumes are customary, and achieving normocapnia is no longer required per se. This is the case for virtually all ventilated patients. One exception, however, is the patient with brain injury and relatively normal lungs, in whom a tight PaCO_2 control is required to avoid undesirable episodes of brain ischemia or hyperemia.

In patients with COPD, data indicate that the quotient between tidal volume and expiratory time—mean expiratory flow—is the principal ventilator setting influencing the degree of dynamic hyperinflation.^{94,100} An arterial oxygen saturation of approximately 90% is sufficient and is usually achieved with moderate oxygen concentrations. A respiratory rate of 12 breaths/min, tidal volume of approximately 8 mL/kg or lower, and a constant inspiratory flow rate of between 60 and 90 L/min are usually acceptable initial settings. These settings need to be readjusted, as needed, once basic respiratory system mechanics and arterial blood gases have been measured. In these patients, the goal is to keep a balance between minimizing dynamic hyperinflation and providing sufficient alveolar ventilation to maintain arterial pH near the low-normal limit, not a

normal PaCO_2 . When patients are receiving ACV and mechanical breaths are triggered by the patient, external PEEP counterbalances the elastic mechanical load induced by intrinsic PEEP secondary to expiratory flow limitation and decreases the breathing workload markedly.¹⁰¹

The ventilator strategy in acute asthma favors moderate tidal volumes, high inspiratory flow rates, and a long expiratory time.^{102–108} These settings avoid large end-inspiratory lung volumes, thus decreasing the risks of barotrauma and hypotension. The main goal in asthma is to avoid these complications rather than to achieve normocapnia. A reasonable recommendation from physiologic and clinical viewpoints when initiating ACV is to provide an inspiratory flow of 80 to 100 L/min and a tidal volume of approximately 8 mL/kg, and to avoid end-inspiratory plateau airway pressures higher than 30 cm H_2O . The respiratory rate should be adjusted to relatively low frequencies (approximately 10 to 12 cycles/min) so as to minimize hyperinflation. These settings are accompanied most often by hypercapnia and respiratory acidosis and require adequate sedation, even neuromuscular blockade in some patients. Ventilator settings should be readjusted in accordance with the time course of changes in gas exchange and respiratory system mechanics.

Most patients with ARDS require mechanical ventilation during their illness. In this setting, mechanical ventilation is harmful when delivering high tidal volumes.^{99,100} There is general agreement that end-inspiratory plateau airway pressure should be kept at values no higher than 30 cm H_2O . End-inspiratory plateau airway pressure, however, is a function of tidal volume, total PEEP level, and elastance of both the lung and chest wall. Importantly, patients with ARDS have small lungs with different mechanical characteristics of the lungs and chest wall,^{101,102} and recommending a single combination of tidal volume and PEEP for all patients is not sound. Patients with more compliant lungs possibly can receive somewhat higher tidal volumes and PEEP levels than those delivered to patients with poorly compliant lungs. As in any other disease state, individual titration of tidal volume and PEEP according to underlying physiologic abnormalities and to the time course of the disease seems the most reasonable.¹⁰⁹ Besides, such an approach serves as a control for comparison purposes.

Important Unknowns and the Future

Mechanical ventilation is instituted mainly to improve gas exchange and to decrease respiratory muscle workload. The clinical response to this lifesaving treatment in terms of gas exchange is usually evaluated by means of intermittent arterial blood-gas measurements, continuous pulse oximetry monitoring, and, less often, monitoring end-tidal CO_2 . These measurements provide an objective way to titrate therapy. Although gas exchange is the main function of the lungs, the respiratory system also has a muscular pump that is central to its main purposes. The way we evaluate the function of the respiratory muscles clinically during the course of ACV and patient–ventilator interactions is rudimentary. Knowing how much effort a particular patient is making and how much unloading is to be provided is very difficult to ascertain on clinical grounds. Too much or too low respiratory muscle effort may induce muscle dysfunction, and this eventually could delay ventilator withdrawal.

When ACV is first initiated, the ventilator usually overcomes the total breathing workload. How long the period of respiratory muscle inactivity is to be maintained is unknown. When ACV is triggered by the patient, multiple factors interplay between the patient and the ventilator. Although high levels of assistance decrease the sensation of dyspnea, they also increase the likelihood of wasted inspiratory efforts (see Fig. 6-7). How ACV is adjusted, in particular concerning inspiratory flow rate and tidal volume settings, is a major determinant of its physiologic effects. If the settings are selected inappropriately, these may lead the physician to erroneously interpret that the problem lies with the patient and perhaps administer a sedative agent when, in reality, the patient is simply reacting against improper adjustment of the machine.

When patients are receiving ACV, they are at risk of undergoing periods of underassistance alternating with periods of overassistance. This is so because of the varying ventilatory demands (see Fig. 6-5) and because the mechanical characteristics of the respiratory system also change over time. The frequency of such phenomena and their clinical consequences are

unknown. The effects of permanent monotonous tidal volume delivery, as well as whether or not sighs are to be used in this setting, also remain to be elucidated.

The only way to interpret clinically whether the patient is doing well or not during ACV is to evaluate respiratory rate and the airflow and airway pressure trajectories over time. During patient-triggered ACV, muscle effort can be estimated by superimposing the current and the passive airway pressure trajectories. Airway occlusion pressure is an important component of the airway pressure trajectory during patient-triggered breaths. This variable is a good estimate of the central respiratory drive and is highly correlated with the inspiratory muscle effort. Such measurements would allow clinicians to analyze trends and estimate patient-ventilator interactions objectively. It is surprising that such sound noninvasive monitoring possibilities have yet to be widely implemented, and it is ironic to realize how many new ventilator modes are introduced without having passed rigorous physiologic and clinical evaluations.

Summary and Conclusion

The most widely used ventilator mode in mechanically ventilated patients continues to be ACV. Many of its physiologic effects are well characterized, and it is conceivable that, in the main, its purposes are met. ACV is also very versatile because it offers ventilator support throughout the entire period of mechanical ventilation. As with any other mode, the effects depend on the way ACV is implemented. The necessity to impose a number of fixed settings, in essence, tidal volume and inspiratory flow rate, implies that the respiratory pump may be unloaded suboptimally and contraction of the respiratory muscles may be asynchronous with the ventilator. The clinical consequences of these phenomena are not negligible.

Since its introduction, ACV implementation has undergone considerable changes, and it is presently applied less aggressively than in the past. Thanks to an enormous amount of physiologic and clinically oriented research, we have learned that ACV can be harmful to patients, injuring both the lungs and the respiratory muscles. Future research should help us to deliver ACV in such a manner that a patient's clinical needs are served more optimally.

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